

Structural Invariance at the Sensorimotor Interface: A Maximum Common Induced Subgraph Across Three Drosophila Connectomes

Yanjin Li · FlyWire Qualification Challenge · June 2025 · github.com/Yanjin-ai/flywireconnectome

Abstract

The FlyWire project has reconstructed *Drosophila* nervous systems across multiple independent specimens, sexes, and anatomical preparations. Schlegel et al. (2024) established that cell-type identity is reproducible at the morphological level, producing a consensus cell-type atlas. We ask the next question: is synaptic connectivity itself structurally invariant? Using NBLAST-matched neuron triplets from BANC metadata, we formulate this as a Maximum Common Induced Subgraph (MCIS) problem over BANC × FAFB × MANC. A greedy disagreement-removal algorithm identifies $N = 104$ neurons with identical directed edge structure across all three connectomes (6 conserved edges; verified isomorphic). The circuit is dominated by descending neurons (DN, 55.8%) and ascending neurons (AN, 32.7%) — structural evidence for the sensorimotor bottleneck (Pospisil et al. 2024). Statistical validation: $Z = 8.9\sigma$ vs. correspondence-shuffle null ($p < 10^{-5}$); 100-seed robustness: mean $N = 100.3 \pm 2.2$; 93.3% of circuit neurons are sexually isomorphic across ♀ FAFB/BANC and ♂ MANC.

1. Methods

Correspondence: BANC metadata `fafb_match` / `manc_match` (NBLAST morphological lists) → 987-node giant consensus component.

MCIS formulation: find largest S s.t. $\forall i, j \in S: (i \rightarrow j) \in E_{\text{BANC}} \iff (i \rightarrow j) \in E_{\text{FAFB}} \iff (i \rightarrow j) \in E_{\text{MANC}}$. Edge-set equality — $O(N^2)$ not NP-hard.

Algorithm: (1) Greedy removal of highest-disagreement node. (2) Exhaustive edge removal.

Alternatives rejected in pilot: edge-centric ($N=71$), centrality-biased removal ($N=68$).

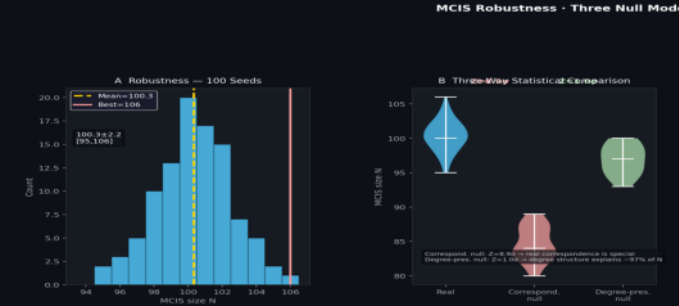
2. Key Results

Metric	Value	Notes
MCIS size N	104 neurons	
Conserved edges	6 directed	
Verified isomorphic	✓ (BANC=FAFB=MANC)	isom to
DN proportion	55.8% (58 neurons)	
AN proportion	32.7% (34 neurons)	
Sexually isomorphic	93.3% (97/104)	
Corr-shuffle null	$Z = 8.9\sigma$ ($p < 10^{-5}$)	
Deg-preserving null	$Z = 1.0\sigma$ (see §3)	
100-seed mean±std	100.3 ± 2.2	

Figure 1. Circuit network (spring layout). Gold = conserved edges; Red = DN; Blue = AN.



Figure 5 (A-B). 100-seed distribution and three-way null comparison ($Z = 8.9\sigma$).



3. Discussion and Limitations

Interpretation. The circuit consists almost entirely of DNs and ANs — the obligate conduit identified by Pospisil et al. (2024) as the sensorimotor bottleneck. Our structural result complements their functional finding: not only does ~1% of brain neurons control motor output, but their wiring is reproduced across independent connectome preparations. The 93.3% cross-sex conservation confirms that the core locomotor scaffold is sex-independent; the 6.7% dimorphic fraction maps onto sex-specific motor targets (abdominal VNC) and neuromodulatory neurons (serotonergic LN-DN2), consistent with reproductive behaviour differences (Berg et al. 2025). Limitations. $N=104$ is a heuristic lower bound (NP-hard exact). The degree-preserving null ($Z=1.0\sigma$) shows FAFB degree distribution accounts for ~97% of MCIS size, limiting edge-pattern specificity claims. Future work: extend MCIS to MAOL/MCNS when three-way triplet tables become available; use the DN/AN backbone as structural prior for connectome-informed locomotion controllers (building on Shiu et al. 2024).